

MATHEMATICS 2270
Quiz 2, Fall semester 2012

Name:.....

1. Find elimination matrices that put $A = [1, 1, 0 | 1, 0, 2 | 2, 2, 6]$ in to an upper triangular form U .

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 2 & 2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 2 & 2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 6 \end{bmatrix} = U$$

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad FEA = U$$

2. Use your answer in 1 to factor $A = LU$ where L is lower triangular.

$$A = E^{-1}F^{-1}U \quad L = E^{-1}F^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

3. For what t is the matrix $A = [1, 1, 1 | 2, t, 0 | 0, 2, 4t]$ invertible?

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & t & 0 \\ 0 & 2 & 4t \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & t-2 & -2 \\ 0 & 2 & 4t \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -2-2t(t-2) \\ 0 & 2 & 4t \end{bmatrix}$$

3 non-zero pivots if $-2-2t(t-2) \neq 0$ i.e. if $1+t^2-2t \neq 0$

i.e. if $t \neq 1$

check $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$ not invertible